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Q. 1 Do as directed

- (i) The number of significant digit(s) in 0.00006032 is/are -----
- (ii) Biased data are considered to be inaccurate/imprecise.
- (iii) Newton Raphson method is an open/close root finding method.
- (iv) The length and breadth of a rectangular sheet of metal are 4.234 and 1.005 cm respectively. The area of the sheet to correct significant figures -----.
- (v) True error is always ----- than estimated error.
- (vi) In Gauss Jordan method for solving linear simultaneous equations, the elimination steps results in ----- matrix.
- (vii) 3/8 Simpson's rule is ----- order accurate.
- (viii) Heun's method for solving ODE is ----- order accurate.
- (ix) IF THEN ELSE is the syntax of Fortran/C computer language.
- (x) While declaring any variable as double precision in FORTRAN code, by default it is considered as real variable (TRUE/FALSE).

Q. 2 (a) Recorded time by the Stopped clock are the most precise data. Justify this statement. (3)

(b) A physical quantity P is related to four observations a , b , c , and d as follows.

$$P = a^3 b^2 / \sqrt{c} d$$

The percentage errors of measurement in a , b , c , and d are 1%, 3%, 4% and 2%, respectively. What is the percentage error in the quantity P . (6)

(c) If $X = 0.51$ and is corrected to 2 decimal places. Find the relative error. (3)

Q.3 (a) In environmental engineering, the following equation can be used to compute the oxygen level in a river downstream from a sewage discharge:

$$c = 10 - 20(e^{-0.2x} - e^{-0.75x})$$

where x is the distance downstream in kilometres. Determine the distance downstream where the oxygen level first falls to a reading of 5. Determine your answer to a 1% error. Initial gauss can be chosen as 1. (8)

(b) Explain the concept of bisection method graphically and write the necessary conditions to find out the real root by bisection method. (4)

Q.4 (a) Explain the concept of Pivoting. (3)

(b) Given the system

$$\begin{aligned}x_1 + x_2 - x_3 &= -3 \\6x_1 + 2x_2 + 2x_3 &= 2 \\-3x_1 + 4x_2 + x_3 &= 1\end{aligned}$$

Solve by Gauss Jordan Method. Show all the steps of computation. (9)

Q. 5 (a) The heat flux q is the quantity of heat flowing through a unit area of a material per unit time. It can be computed with Fourier's law $q = -k \frac{dT}{dx}$ where q has unit W/m^2 and k is a coefficient of thermal conductivity that parameterizes the heat conducting properties of the material and has units of $W/^\circ C \cdot m$, T = Temperature ($^\circ C$); and x = distance (m) along the path of heat flow. The following temperature are measured from the surface ($x = 0$) into a stone wall

x (m)	0	0.1	0.2
T ($^\circ C$)	20	17	15

If the flux at $x = 0$ is $60 W/m^2$, compute k . (8)

(b) Derive the multiple application Trapezoidal formula, to be used for numerical integration. (4)

Q. 6 (a) The table below gives the temperature T ($^\circ C$) and length l (mm) of a heated rod. If $l = a_0 + a_1 T$, find the values of a_0 and a_1 using linear least squares

T	40	50	60	70	80
l	600.5	600.6	600.8	600.9	601.0

(8)

(b) Write the algorithm of Bisection method. (4)

Q. 7 (a) Explain the concept of Heun's Method and write the necessary expression of Predictor and Corrector. (4)

(b) Solve the following ODE using Euler's method from $t = 0$ to 3 taking step size of 0.5. Find the local error in each step.

$$\frac{dy}{dt} = -y + t \quad \text{given } y(0) = 1 \quad (8)$$

Q. 1 Do as directed

- (i) The number of significant digit(s) in 0.0500 is/are -----
- (ii) Measurements done by the scale having calibration error will be Biased/Uncontained.
- (iii) Truncation error in the central difference finite divide difference is of the order of-----.
- (iv) Sum of 2.29 and 62.7 giving respect to accuracy level of each number is -----
- (v) Estimated error is always ----- than true error.
- (vi) Well conditioned systems are those where small changes in coefficients result in ----- changes in solution.
- (vii) 1/3 Simpson's rule is ----- order accurate.
- (viii) Solution of the cubic differential equation with RK third order method will be -- -----.
- (ix) The valid Fortron code for $e^{\tan^3(a)}$ is -----
- (x) While declaring any variable as doubleprecision in FORTRON code, by default it is considered as real variable (TRUE/FALSE).

Q. 2 (a) What do you understand by Accuracy and precision. Explain with suitable example. (4)

(b) Evaluate the sum $S = \sqrt{3} + \sqrt{5} + \sqrt{7}$ to 4 significant digits and find its absolute and relative error. (5)

(c) Round off the following to four significant digits. (3)
363042, 0.0063945, 0.090038

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$$c = 10 - 20(e^{-0.2x} - e^{-0.75x})$$

where x is the distance downstream in kilometres. Determine the distance downstream where the oxygen level first falls to a reading of 5. Determine your answer to a 1% error. Initial gauss can be chosen as 1. (8)

(b) Explain the concept of bisection method graphically and write the necessary conditions to find out the real root by bisection method. (4)

Q.4 (a) What do you understand by "Ill conditioned" and "well conditioned". (3)

(b) Given the system

$$\begin{aligned} x_1 + x_2 - x_3 &= -3 \\ 6x_1 + 2x_2 + 2x_3 &= 2 \\ -3x_1 + 4x_2 + x_3 &= 1 \end{aligned}$$

Solve by naïve Gauss elimination with partial pivoting. Show all the steps of computation. (9)

- Q. 5 (a) The heat flux q is the quantity of heat flowing through a unit area of a material per unit time. It can be computed with Fourier's law $q = -k \frac{dT}{dx}$ where q has unit W/m^2 and k is a coefficient of thermal conductivity that parameterizes the heat conducting properties of the material and has units of $W/^\circ C \cdot m$, T = Temperature ($^\circ C$); and x = distance (m) along the path of heat flow. The following temperature are measured from the surface ($x = 0$) into a stone wall

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- (b) Derive the multiple application Trapezoidal formula, to be used for numerical integration.

(4)

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(8)

T	40	50	60	70	80
l	600.5	600.6	600.8	600.9	601.0

- (b) Write the algorithm of finding the root by bisection method.

(4)

- Q. 7 Solve the following ODE using Euler's method from $t = 0$ to 3 taking step size of 0.5. Find the local error in each step.

$$\frac{dy}{dt} = -y + t$$

$$\text{given } y(0) = 1$$

(12)
